

Breast Ultrasound Image Classification

1. Methodology

1.1 Dataset & Train/Test Split

- **Source:** BUSI dataset (sabahezaraki/breast-ultrasound-images-dataset) downloaded via kagglehub.
- **Mask exclusion:** Images with filenames containing "_mask" are filtered out using `is_valid_file()`.
- **Split:** 80/20 stratified train-test split, `random_state = 42` (SEED). Test set = 156 samples (benign: 87, malignant: 42, normal: 27).
- **Reproducibility:** All random seeds fixed (Python random, NumPy, PyTorch CPU/CUDA). `torch.backends.cudnn.deterministic = True`.

1.2 Handling Class Imbalance

- **Oversampling:** Minority classes (malignant = class 1, normal = class 2) are oversampled in the training set at a 1.0× rate (100% of their training count), adding 274 augmented extra samples. Benign (class 0) is not oversampled.
- **Alpha-weighted Focal Loss:** Per-class alpha = [1.0, 1.2, 2.0] (benign, malignant, normal) combined with $\gamma = 2$. This down-weights easy well-classified examples and focuses training on hard minority cases.

1.3 Data Augmentation

- **Pipeline (augmented copies only):** Resize → 224×224, RandomHorizontalFlip, RandomRotation($\pm 15^\circ$), ToTensor, Gaussian noise $x + 0.05 \times N(0,1)$.
- **Original samples:** Resize → 224×224, ToTensor only — no augmentation applied.
- **Note:** No Normalize() transform is applied in this version. Pixel values remain in [0, 1] (plus small noise for augmented copies).

1.4 Model Architecture — Custom CNN (422,659 parameters)

Block	Output shape	Operations	Notes
Block 1	112×112×32	Conv2d(3 → 32, 3×3, p=1), BatchNorm2d, ReLU, MaxPool2d(2)	
Block 2	56×56×64	Conv2d(32 → 64, 3×3, p=1), BatchNorm2d, ReLU, MaxPool2d(2)	
Block 3	28×28×128	Conv2d(64 → 128, 3×3, p=1), BatchNorm2d, ReLU, MaxPool2d(2)	
Block 4	1×1×256	Conv2d(128 → 256, 3×3, p=1), BatchNorm2d, ReLU, AdaptiveAvgPool2d(1,1)	Global avg pool
Classifier	3	Flatten, Linear(256 → 128), ReLU, Dropout(0.4), Linear(128 → 3)	

1.5 Training Configuration

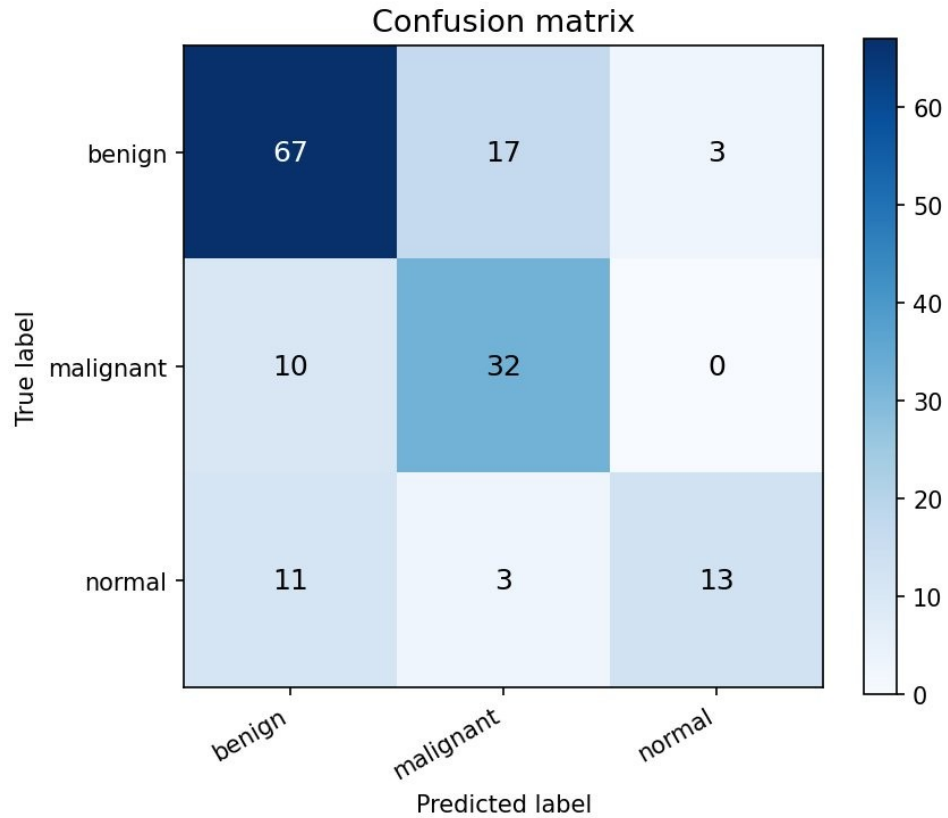
Setting	Value
Optimiser	Adam ($\text{lr} = 1 \times 10^{-4}$)
Loss	Focal Loss ($\gamma = 2$, $\alpha = [1.0, 1.2, 2.0]$ for benign / malignant / normal)
Epochs	15
Batch size	32
Input size	$224 \times 224 \times 3$ (RGB)
Train batches / Test batches	29 / 5
Augmented extra samples	274 (malignant $\times$ 1.0 + normal $\times$ 1.0)

2. Results

2.1 Per-Class Classification Report

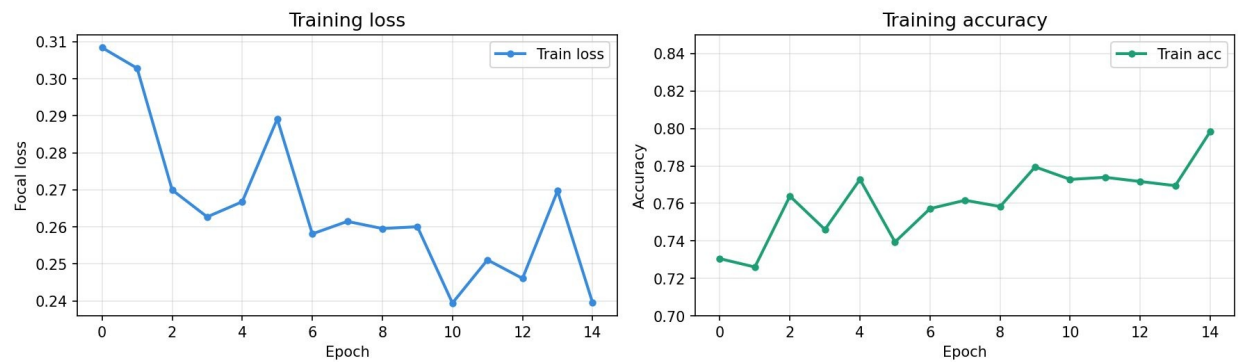
Class	Precision	Recall	F1-Score	Support
Benign	0.76	0.77	0.77	87
Malignant	0.62	0.76	0.68	42
Normal	0.81	0.48	0.60	27
Overall accuracy	—	—	0.72	156
ROC-AUC (macro OvR)	—	—	0.868	—

2.2 Confusion Matrix

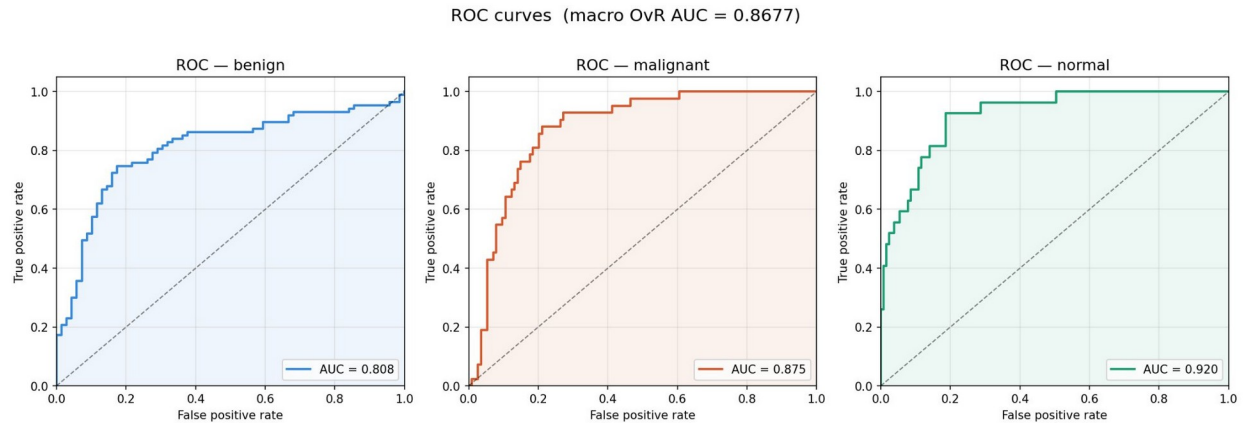


2.3 Training Curves

Training curves



2.4 ROC Curves



3. Key Observations

- **Overall accuracy 72%, ROC-AUC 0.868** — the model discriminates well at the probability level even though hard-label accuracy is moderate.
- **Benign (recall 77%):** 17 cases wrongly predicted as malignant, 3 as normal. Largest class and hardest boundary.
- **Malignant (recall 76%):** 10 misclassified as benign; 0 confused with normal — the clinically critical benign $\leftrightarrow$ malignant boundary is partially controlled.
- **Normal (recall 48%):** Weakest class. 11 of 27 normal samples predicted as benign. Likely caused by visual overlap and the smallest support (27 test samples).
- **Training instability:** Loss and accuracy oscillate across epochs (no LR scheduler, no Normalize() transform). Adding CosineAnnealingLR and normalisation are the immediate next steps.
- **No input normalisation:** The normal_transform pipeline omits Normalize(). This slows convergence as pixel values are not zero-centred for the conv layers.